**CSE 4283 / 6283 – Software Testing and Quality Assurance**
**Assignment-3, 100 Points**

## Objective

Understand and practice unit testing.

## Activities

### Step 0: Set up the environment

Download and install Jetbrains Rider. If you need help, follow the instructions in the following video and set up the environment required for this assignment.
https://www.dropbox.com/scl/fi/vgukxyih4cb8dpbv64dl7/2025-01-22-15-59-19.mp4?rlkey=qrz09b60yfhdo5qp2odnc4cu9&e=1&dl=0

You can also find the recording on the course website under "Instructional Recordings" in "Modules".

### Step 1: Watch these videos:

1. Unit testing:

   https://youtu.be/HYrXogLj7vg?si=glGGfnzWjCMvA0Bk

2. Technical Debt & Refactoring:

   https://youtu.be/xbGVq0VkCRg?si=QSWSUX165AFs82Pp

3. C# Basics:

https://youtu.be/ZqDtPFrUonQ?si=X9aSWYEl8qoknCyM

https://youtu.be/q7aWkjH3UUI?si=hAqzix58VruzMhAQ

https://youtu.be/4_J_CLKwoto?si=J2DDC21rNtAg9Xyv

https://youtu.be/vpT4-7YdV7M?si=k_qClzdbqFWIy6tk

https://youtu.be/56bbjE4assU?si=MSWYcoJsl9MszE8p

https://youtu.be/_TxOmmJTd98?si=cCtChPU4LwOPfUuj

https://youtu.be/aQ8YkJrAbzE?si=dyLRyPEoTPKMnG-l

https://youtu.be/gyal6TbgmSU?si=asy-5X4zeqX75wmf

**Step 2: Extract the Project Files**

1. Locate the provided ZIP file:

   Shopping.zip

2. Extract the ZIP file to a suitable location on your computer.

3. Open the extracted folder and locate the solution file:

   Shopping.sln

**Step 3: Open the Project in Rider IDE**

1. Launch JetBrains Rider.

2. In Rider, select Open and navigate to the extracted folder.

3. Open Shopping.sln to load the project.

**Step 4: Run the Existing Unit Test**

1. In Rider, open the Test Explorer (or right-click inside ShoppingCartTests.cs).

2. Run the existing unit test.

3. Confirm that the test passes successfully.

**Step 5: Review the Existing Code**

1. Using the Solution Explorer in Rider, locate and open the class:

   ShoppingCart.cs

   Review the implementation of the shopping cart logic.

2. Locate and open the test file:

   ShoppingCartTests.cs

   Review the structure and implementation of the existing unit test.

**Step 6: Complete the Programming Tasks**

1. Write Unit Tests for RemoveItem
    - Ensure that removing an existing item decreases the item count.
    - Ensure that removing a non-existing item does not change the item count.
2. Write Unit Tests for GetTotal
    - Add multiple items to the cart and verify that the total price is correct.
    - Ensure that the total price is zero when the cart is empty.
3. Write Additional Tests for Edge Cases
    - Test adding duplicate items to the cart.
    - Test removing an item from an empty cart.
    - Test handling negative or zero prices when adding items.
4. Extend the ShoppingCart Class
    - Implement a new method that returns a list of all items in the cart.
    - Write unit tests to verify the functionality of this method.

**Step 7: Submit Your Work**

1. Ensure all tests pass successfully in Rider.
2. Compress the entire project folder (the folder containing Shopping.sln) into a ZIP file.
3. Upload the ZIP file to Canvas for submission.